

Program-3

Develop a program to implement Principal Component Analysis (PCA) for reducing the dimensionality of the Iris dataset from 4 features to 2.

Theory

Principal Component Analysis (PCA)

Principal Component Analysis (PCA) is a statistical dimensionality reduction technique used to reduce the number of features in a dataset while preserving most of the important information or variance present in the data.

In many datasets, there may be a large number of correlated features. PCA transforms the original features into a new set of uncorrelated variables called **Principal Components**. These principal components are arranged in such a way that:

- The first principal component captures the maximum variance in the data.
- The second principal component captures the next highest variance.
- Remaining components capture lesser variance.

PCA helps in:

- Reducing computational complexity
 - Removing redundancy
 - Improving visualization
 - Reducing noise in data
-

Working of PCA

The PCA process involves the following steps:

1. Standardize the dataset.
2. Compute the covariance matrix.
3. Calculate eigenvalues and eigenvectors.
4. Select the principal components with highest eigenvalues.
5. Transform the original dataset into a lower-dimensional dataset.

The transformation is represented as:

$$Z = XW$$

Where:

- (X) = Original data matrix
- (W) = Matrix of principal components (eigenvectors)
- (Z) = Transformed lower-dimensional data

Iris Dataset

The Iris dataset is a well-known dataset in machine learning containing:

- 150 samples of iris flowers
- 4 numerical features:
 - Sepal Length
 - Sepal Width
 - Petal Length
 - Petal Width
- 3 flower classes:
 - Setosa
 - Versicolor
 - Virginica

In this experiment, PCA is applied to reduce the 4-dimensional feature space into 2 principal components for easier visualization and analysis.

Advantages of PCA

- Reduces dimensionality of data
- Improves visualization
- Removes correlated features
- Reduces storage space
- Speeds up machine learning algorithms

Applications of PCA

- Image processing
- Face recognition
- Data compression
- Pattern recognition
- Machine learning preprocessing

```
[1]: import pandas as pd
df = pd.read_csv("iris.csv")
df.head()
```

```
[1]:
```

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

```
[2]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.datasets import load_iris
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
```

```
[3]: # Load dataset
iris = load_iris()

# Features (X) and target (y)
X = iris.data
y = iris.target

print("Shape of X:", X.shape)
```

Shape of X: (150, 4)

```
[4]: scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

```
[5]: pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)

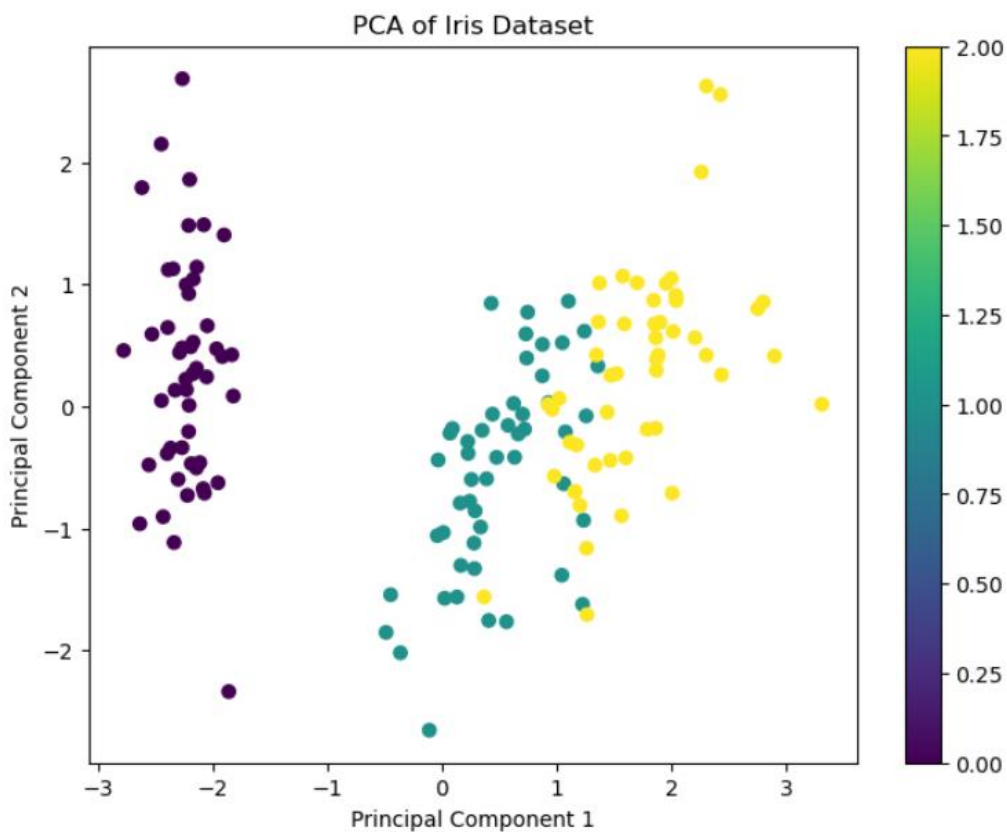
print("New shape:", X_pca.shape)
```

New shape: (150, 2)

```
[6]: plt.figure(figsize=(8,6))

scatter = plt.scatter(
    X_pca[:, 0],
    X_pca[:, 1],
    c=y,
    cmap='viridis'
)

plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.title("PCA of Iris Dataset")
plt.colorbar(scatter)
plt.show()
```



Result

Thus, Principal Component Analysis (PCA) was successfully implemented on the Iris dataset to reduce the dimensionality from 4 features to 2 principal components while preserving maximum variance in the data.